

SUPRIYA M
Data Engineer

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PROFILE SUMMARY:

- Accomplished **Data Engineer** with 10+ years of expertise in architecting scalable data infrastructures, optimizing pipelines and delivering actionable insights to drive business success.
- Extensive expertise in **ETL/ELT** development, advanced data modeling, and optimizing scalable data pipelines for high-performance analytics, with strong background in designing and implementing relational databases, ensuring data integrity and normalization.
- Advanced proficiency in cloud platforms, including **AWS (Redshift, EMR, Glue)**, **Azure**, and **Google Cloud**, for deploying and managing robust data solutions.
- Designed and implemented scalable ETL pipelines leveraging AWS services (S3, Glue, Lambda, EMR, Redshift) for cloud-based data processing.
- Developed data workflows and automation scripts using Python; implemented unit and integration tests with Pytest to ensure data quality.
- Utilized Apache Spark for distributed data processing and optimization of large-scale datasets.
- Applied advanced SQL techniques for querying, transformation, and management of relational databases.
- Built and maintained data warehousing solutions with strong focus on ETL design, performance, and scalability.
- Collaborated in Agile teams using CI/CD pipelines and Git for version control to streamline development and deployment.
- Proficient in **Python**, **Airflow**, and DAG orchestration, with hands-on experience in managing **CI/CD pipelines** and **DevOps practices**.
- I am skilled in handling **unstructured data formats** like Parquet, JSON, AVRO, and XML.
- Proficient in data modeling with **Flowcharts** and **Screen layouts**, ensuring effective communication and smooth implementation of scalable data systems.
- Advanced knowledge of Google Cloud Platform (Cloud Run, Cloud SQL, BigQuery, **Cloud Functions**, **Compute Engine**, **Cloud Storage**, **Container Registry API**) with expertise in security best practices.
- Proficient in **containerization (Docker)** and **orchestration (Kubernetes)** for deployment, scaling, and maintenance of cloud-native applications.
- In-depth expertise in relational and **NoSQL** databases, including **PostgreSQL**, **MySQL**, **MongoDB**, and **Cassandra**, with a focus on optimizing data architecture, enhancing query performance, and ensuring efficient data storage.
- Strong understanding of **Hadoop v1/v2 architecture**, with hands-on experience in components such as Job Tracker, Task Tracker, Name Node, Secondary Name Node, Data Node, and YARN architecture.
- Coordinated and managed the migration of complex data flows, including the transition from legacy systems (**Teradata**) to modern data platforms such as **Google Cloud Platform**, ensuring high data integrity and seamless performance.

PROFESSIONAL EXPERIENCE:

Client: Safelite Auto glass
Senior Data Engineer:

Columbus, OH
August 2023 – Present

Project: Automated Data Pipeline Orchestration with Apache Airflow |*Apache Airflow, Kafka, Snowflake, AWS, Jenkins, Docker, Prometheus, Grafana, Slack*

Architected and deployed scalable data pipelines using Apache Airflow, Kafka, Snowflake, and Big Query to automate end-to-end data workflows and improve system reliability. Directed a successful migration from Teradata and implemented CI/CD practices with Jenkins and Terraform. Enhanced observability and system diagnostics using Prometheus and Grafana, resulting in improved platform performance and streamlined release cycles.

- Architected and deployed scalable data pipelines using **Snowflake** as a high-performance cloud data warehouse to automate end-to-end data workflows and improve system reliability.
- Experience in managing **batch and real-time workflows**, ensuring **data integrity, reliability, and automation** in cloud-based environments.
- Supervised end-to-end architecture and development of large-scale data integration solutions, including Batch and Stream processing methodologies, using tools **Snowflake, Big Query, Redshift, Kafka, and Airflow**.
- Integrated EHR systems to streamline patient data management, ensuring smooth interoperability across healthcare platforms using HL7 standards for secure, compliant data exchange improved data accessibil

- Designed detailed flowcharts and screen layouts to visualize complex data flows and system architectures, enhancing team communication and ensuring seamless implementation of data solutions.
- Integrated **Google Cloud Security Scanner** into CI/CD workflows, ensuring proactive remediation and compliance.
- Developed **API integrations with GCP services** (Cloud Run, Cloud SQL, BigQuery, Cloud Functions) to enable seamless data ingestion and reporting.
- Designed and developed **Azure Data Factory (ADF) pipelines** for seamless data ingestion and transformation across various sources.
- Implemented and managed **CI/CD pipelines in Azure DevOps**, ensuring automated deployment, monitoring, and governance of data integration workflows.
- Applied **TDD** to ensure error-free code and high-quality **data pipelines**, improving the reliability of **ETL processes** and reducing bugs in production. Implemented unit and integration testing with Pytest to validate ETL workflows and improve pipeline reliability.
- Supervised end-to-end architecture and development of large-scale data integration solutions, including Batch and Stream processing methodologies, using a combination of **Snowflake**, Big Query, Redshift, Kafka, and Airflow.
- Actively participated in **Agile/Scrum** methodologies, contributing to daily standups, sprint planning, sprint reviews, and retrospectives, ensuring iterative and incremental progress on data engineering tasks.
- Architected BI solutions on Azure using Data Lake and Data Factory and developed **Scala** batch scripts with Spark to transform data from **AWS S3**.

Environment: Apache Airflow, Apache Nifi, AWS Glue, Apache Spark, Kafka, Databricks, PySpark, CI/CD Snowflake, Big Query, HER, Scala, RDS, C++, C#, Talend, Informatica, Kibanna, AWS, S3, Agile/Scrum, GCP, Azure, Docker, Kubernetes, Terraform, Ansible, Prometheus, Grafana, Git, Jenkins, GitLab CI

Client: EMC Insurance

Des Moines, Iowa

Data Engineer:

April 2021 – July 2023

Project: Scalable Data Pipeline on Azure | *SQL, AWS, Azure Synapse Analytics, ADF, Python, Kubernetes, CloudWatch, Azure Blob Storage*

Developed and implemented an end-to-end data pipeline utilizing Azure Synapse, Azure Data Factory (ADF), and Databricks to streamline ETL processes and automate data ingestion from a variety of structured and semi structured formats, including JSON, XML, and AVRO. Optimized SQL queries to enhance processing efficiency and introduced real-time monitoring and SLA tracking to improve pipeline reliability and system transparency. Additionally, automated deployment workflows by integrating CI/CD pipelines with Git, ensuring consistent and efficient delivery across environments.

- Designed, built, and optimized large-scale data pipelines using **batch and real-time processing frameworks** (e.g., Airflow, Kafka, Spark, Dataflow).
- Integrated PowerShell scripts into Azure Data Factory (ADF) workflows for automated system-level checks and resource validation.
- Developed and optimized data pipelines for seamless integration and transformation, handling both batch and stream processing methodologies using **SQL, Python**, and tools like **Five Tran, Snowflake**, and **Redshift**.
- Developed and implemented an end-to-end data pipeline utilizing **Azure Synapse, Azure Data Factory (ADF)**, and Databricks to streamline ETL processes and automate data ingestion from a variety of structured and semi-structured formats.
- Applied **Test-Driven Development (TDD)** principles to ensure robust, error-free code in **ETL processes**, significantly enhancing pipeline reliability, and minimizing system downtime.
- Developed automation scripts using Python to validate performance of ETL tools across multiple platforms, improving testing speed and reliability.
- Extensive experience with **GCP (Big Query, Dataflow)**, **AWS (Redshift, Glue)**, and **Snowflake** for scalable data architecture.
- Built and optimized **BigQuery pipelines on GCP**, integrating with **Dataflow** for batch and stream processing.
- Containerized ETL workloads with **Docker** and deployed them on **Kubernetes**, improving scalability and maintainability.
- Implemented **IAM role-based access controls (RBAC) in GCP** to strengthen platform security.
- Collaborated with system engineering teams to analyze processor-level performance metrics and translate findings into actionable optimizations for ETL job design.
- Built custom monitoring dashboards in **Power BI** to visualize tool efficiency, CPU usage, and resource bottlenecks across Azure-hosted pipelines.
- Built custom monitoring dashboards in Power BI to visualize tool efficiency and resource bottlenecks across **Azure-hosted pipelines**.
- Skilled in using **Azure SQL Database, Blob Storage, Data Lake**, and **Synapse Analytics** for managing and processing large-scale datasets.

Environment: Apache Airflow, Luigi, Prefect, Elastic Search, SDLC, Azure, SQL (PostgreSQL, MySQL), NoSQL (MongoDB, Cassandra), Big Query, Data Lakes, CMM/CMMI3, Apache Flink, GIT, Apache Beam, Informatica, Apache Nifi, JSON, CSV, Sqoop, Kubernetes, GCP

Client: Comerica Bank

Dallas, TX

Data Engineer I:

January 2019- March 2021

Project: Real-Time Data Pipeline for Stock Market Analysis | *GCP, TDD, Kinesis, S3, Apache Kafka, Google Pub. Apache Airflow, Big Query, dtl, Terraform, GitHub, Power BI, Tableau*

Designed, developed, and maintained scalable real-time and batch data pipelines leveraging Kafka, Pub/Sub, Apache Airflow, and cloud-based data warehouses including Big Query and Snowflake. Streamlined transformation logic using dbt and automated infrastructure provisioning and deployment through Terraform integrated with CI/CD workflows. Delivered low-latency data processing and enhanced reliability by implementing proactive monitoring, validation mechanisms, and automated data quality checks.

- Supported development and maintenance of ETL (Extract, Transform, Load) pipelines to automate the movement and transformation of data from source systems to data warehouses or data lakes.
- Worked with the **Hadoop ecosystem**, including data processing, storage, and management within cloud environments, with a strong emphasis on **data visualization, information architecture**, and ensuring the highest levels of **data quality**.
- Demonstrated extensive expertise in utilizing **AWS services** such as **S3, EC2, EMR, Lambda, Kinesis, Redshift, RDS, DynamoDB**, and others to drive efficient cloud infrastructure management and deliver robust data solutions.
- Automated data workflows with Python and CI/CD pipelines to improve deployment efficiency.
- Skilled in using Azure Databricks for building scalable data workflows, and proficient with Azure's security tools like Active Directory and Key Vault for secure data operations.
- Expertise in **writing optimized SQL queries** and working with NoSQL databases like **MongoDB or Cassandra**.
- Utilized **Azure Databricks** for building scalable data workflows and was proficient with Azure's security tools like Active Directory and Key Vault for secure data operations.
- Monitored and optimized pipeline performance using AWS CloudWatch, resolving bottlenecks and automating ETL tasks with **AWS Glue**.
- Integrated **security scanning reports** (Google Cloud Security Scanner, Tenable) into GitHub workflows, ensuring proactive remediation.
- Developed and maintained dashboards in **Tableau** and **Power BI** to visualize key metrics and performance indicators, supporting business intelligence efforts.
- Created SQL queries for efficient data extraction, aggregation, and reporting from relational databases.
- Implemented **CI/CD** pipelines for Airflow **DAGs** using **GitHub Actions & Terraform**, reducing deployment **time by 70%**. Established Airflow monitoring dashboards (**Grafana, Prometheus**), improving failure detection & debugging efficiently.
- Monitor the health and performance of data storage solutions (e.g., **GCS, Amazon S3, Azure Blob Storage**) to ensure data is securely stored and accessible for analytics.
- Designed, developed, and maintained scalable real-time and batch data pipelines that leveraged cloud-based data warehouses, including **Snowflake**.
- Familiarity with automating data workflows and deployments using **GitHub Actions, Jenkins, or Terraform**.

Environment: SQL, AWS, GCP, ETL, Airflow, Python, S3, EC2, EMR, MongoDB, Cassandra, Lambda, Kinesis, Redshift, RDS, DynamoDB, AWS Glue, AWS Lambda, AWS Cloud Watch, Apache Nifi, Apache Kafka, Azure Blob storage, Airflow DAG

Client: Marathon Oil

Houston, TX

ETL Developer:

February 2015 -December 2018

Project: ETL Pipeline for Sales Data Integration and Analytics| *SQL, Azure (Data Factory, Synapse Analytics, Databricks, Monitor) SQL Database, Apache Spark, Power BI*

Architected and implemented a scalable and reliable automated ETL solution for sales data leveraging SQL and Azure services, including Data Factory, Spark SQL, and Synapse Analytics. Ensured data freshness with daily automated updates and provided critical business insights via real-time Power BI dashboards, with pipeline performance monitored through Azure Monitor.

- Successfully designed, developed, and maintained high-performance ETL pipelines for integrating data across diverse systems, including cloud platforms like **GCP, AWS, and Azure**, ensuring seamless data flow and transformation.

- Designed performance test scenarios using synthetic data loads to stress test Azure ETL pipelines and proactively identify scaling limitations.
- Designed comprehensive **flowcharts** for ETL pipeline architecture, clearly documenting data flow from source to destination, ensuring transparency and understanding among developers and stakeholders.
- Implementing **SQL scripts** to clean, format, and transform raw data. Tasks such as data normalization, deduplication, and applying business logic are done using SQL functions and expressions.
- Utilized Azure SQL Database as a source or destination for relational data in the ETL process. It integrates seamlessly with other Azure services like **Azure Data Factory** for data movement.
- Managed the **ETL service** to automate the extraction, transformation, and loading of data between premises and cloud data sources, integrating various connectors for relational databases, data lakes, and cloud storage.
- Architected and implemented a scalable and reliable automated ETL solution for sales data leveraging **Azure services**, including **Data Factory, Spark SQL, and Synapse Analytics**.
- Developed system-level validation scripts in Python and Bash to verify data integrity and load performance during cloud migration processes.
- Migrated ETL workloads into **Docker containers**, standardizing runtime environments and reducing deployment inconsistencies.
- Automated remediation workflows with **Python and Bash**, ensuring compliance with security policies.
- Gained extensive experience with **cloud platforms** (e.g., **AWS, Google Cloud**) to efficiently manage data storage, processing, and analytics, enhancing scalability and performance across data solutions.
- Integrated data pipelines with analytics tools (**Tableau, Power BI**) and machine learning models, delivering actionable insights and predictive analytics to stakeholders.

Environment: SQL, Apache Nifi, Apache Spark, Airflow, AWS Glue, TDD, Python, Scala, Power BI, Apache Kafka, Spark, Tableau, Power BI, Databricks, HL7, Oracle, PostgreSQL, MySQL, ETL, Athena, Kinesis, RDS, Informatica, Talend, GitLab, Azure, Docker, Grafana, Prometheus, Kibana

TECHNICAL SKILLS:

- **Cloud:** AWS (S3, EC2, Lambda), Azure, Google Cloud Platform
- **Test, Build and Automation:** Jenkins, GitHub Actions, Apache Airflow, Luigi, Prefect
- **Continuous Monitoring:** Datadog, Splunk, ELK Stack, Prometheus, Grafana
- **Bigdata Frameworks:** Apache Spark, Apache Hadoop (HDFS, MapReduce, YARN), Apache Flink
- **Version Control Tools:** Git, GitHub, Gitlab, Bitbucket
- **Scripting Languages:** Python, R, Scala, Bash/Shell, JavaScript (Node.js)
- **Technology Frameworks:** Electronic Health Records (EHR) HL7 solutions Elastic Search engine
- **Databases:** Oracle 11g/10g/9i, SQL, MySQL, NoSQL (MongoDB, DynamoDB, Cassandra)
- **SDLC:** Agile, Waterfall, CI/CD Pipeline, Data Modeling
- **Networking/Protocols:** HTTP, HTTPS, TCP/IP, FTP
- **Security Protocols:** IAM, RBAC, Encryption (SSL/TLS, AES), GDPR, CCPA
- **ETL & Orchestration Tools:** Informatica, Apache Airflow, AWS Glue, Azure Data Factory, Google Dataflow

EDUCATION:

Kent State University Master of Science in Computer Science – GPA: 3.78	Kent, OH
Jawaharlal Nehru Technological University Hyderabad Bachelor of Science - CGPA: 9.0	Hyderabad, Telangana, India